

PHILOSOPHY OF SCIENCE

LESSON 9

PHILOSOPHICAL ASSUMPTIONS IN SCIENCE

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SUMMING UP: NELSON 2016

BUSINESS: A SCIENCE? III THE WORLD(S) WE STUDY

1) "First, while the subject matter they study falls into different classes, within each class there is considerable **heterogeneity**, which cannot be reduced to a small number of well defined variables, given the questions being asked of that subject matter. And in many cases the classes themselves do not have sharply defined **boundaries**" (Nelson 2016: p. 1697)

2) "Second, the forces and conditions that **influence** the subjects of study in ways that the science seeks to understand are **numerous**, highly **variable**, and often cannot be **separated** sharply one from another. Indeed the circumstances associated with any particular set of observations may need to be understood as, in some sense, **unique**." (p. 1697)

3) "...the subject matter they study often **changes over time**..." (p. 1698) – see also lesson 9



The world limits what science we can do (p. 1693)
Kuhn focused not on the world, but human/social factors
Popper focused on how nature of induction/deduction limits

THE CLAIM AND ARGUMENT, BRIEFLY PUT

1. Introduction

Should the social sciences follow the standards and approach of physics?

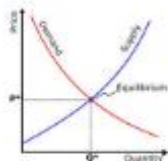
In a nutshell, the argument of this essay is this. For centuries Newtonian (and later post Newtonian) physics has been regarded by both scholars who analyze science, and laypersons impressed with the accomplishments of the scientific enterprise, as the model of what a science should be. It has been the hope, and the expectation, that if sufficient resources and talent were put into the sciences concerned with other phenomena – in particular the life sciences and the behavioral and social sciences – the kind of deep and broad and precise knowledge that had been attained in the physical sciences could be attained there too, and with that knowledge the ability to design technologies, ways of doing things more generally, that would be as effective in these arenas as the technologies made possible by physics. But this is an illusion. The subject matter of the other sciences is different, and the differences matter in terms both of the kind of knowledge and power one can expect from research in these fields, and the



BUSINESS: A SCIENCE? IV NELSON 2016 ON NATURE OF SOCIAL SCIENCE CONSTRUCTS (MEASURES OF CONCEPTS)



The speed of light =
299 792 458 m / s



"I want to suggest that, where the underlying conception of the phenomenon being studied has **blurry edges**, a **constructed number** meant to measure an aspect of it may take on a definitional quality. That is, the number, and the assumptions behind its **construction**, comes to be treated as **what the phenomenon is**, rather than as a **useful if limited measure of that phenomenon**. I propose that this is a real danger, and needs to be guarded against." (Nelson 2016: p. 1695, my emphasis)



Ideas about 'science' influence

- How to do and report science
- What problems to study
- What science to fund

AGENDA

Philosophical assumptions in science	Guba 1990
i) Positivism, post-positivism and constructivism ii) Implications of different philosophical assumptions iii) Final perspectives on theory	

In order to understand Guba 1990, we need to understand two interrelated sets of concepts

- Set 1: Ontology, epistemology, methodology (Guba's paradigms ≠ Kuhn's paradigms)
- Set 2: Positivism, post-positivism, constructivism

Explained via in-depth reading of Guba 1990



"Guba's paradigm: **A basic set of beliefs that guides action**, whether of the everyday garden variety or action taken in connection with a disciplined inquiry." (Guba, 1990, p. 17)

TERMINOLOGY: ONTOLOGY, EPISTEMOLOGY & METHODOLOGY

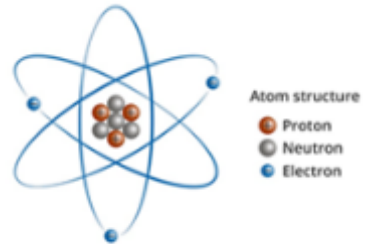
Guba 1990 uses these concepts – but what do they mean?
What question are they even addressing?

- (1) *Ontological*: What is the nature of the “knowable”? Or, what is the nature of “reality”?
- (2) *Epistemological*: What is the nature of the relationship between the knower (the inquirer) and the known (or knowable)?
- (3) *Methodological*: How should the inquirer go about finding out knowledge?

Guba 1990 p. 18

See 2021 video here: <https://youtu.be/37-4-0jlf7k>

WHAT WORLD (ONTOLOGY) ARE WE THEORIZING ABOUT AND HOW DO WE ACCESS IT/ KNOW (EPISTEMOLOGY)? II



- (1) *Ontological*: What is the nature of the “knowable”? Or, what is the nature of “reality”?
- (2) *Epistemological*: What is the nature of the relationship between the knower (the inquirer) and the known (or knowable)?
- (3) *Methodological*: How should the inquirer go about finding out knowledge?

Guba 1990 p. 18

Classical physics

- ‘See’ the stuff we investigate
- Uncover laws that govern reality



We can't directly 'see' business administration concepts

"YES, PRIME MINISTER" I



<https://www.youtube.com/watch?v=G0ZZJXw4MTA>

"YES, PRIME MINISTER": IS THE PROBLEM ONTOLOGICAL OR EPISTEMOLOGICAL, OR BOTH?

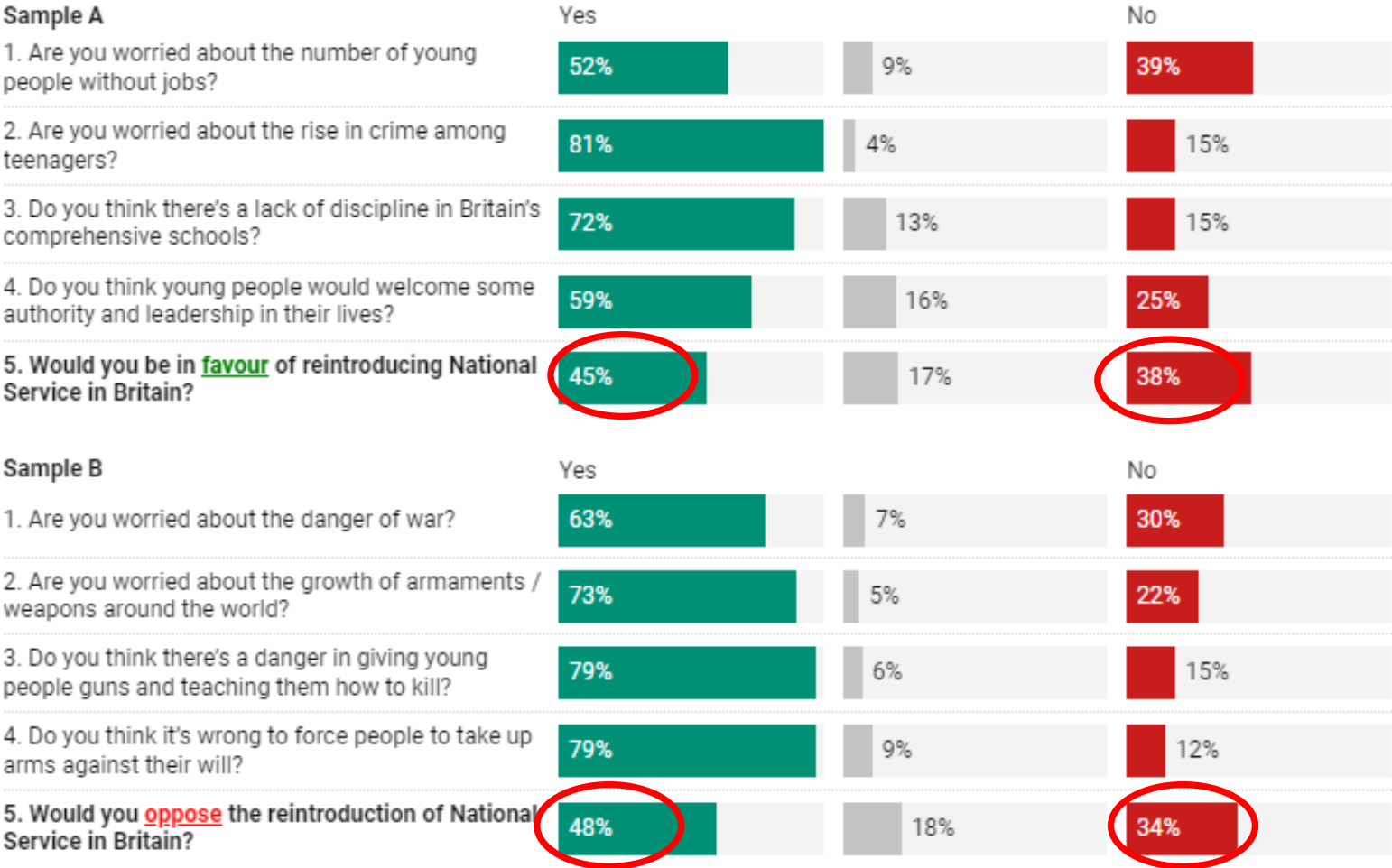


- Ontology: Does an independent opinion on national service exist?
- Epistemology: Is it because we don't have access to what people actually think?

How one tries to
obtain
knowledge
(asks) matters!



Yes, Prime Minister, question ordering does matter...



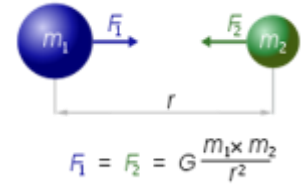
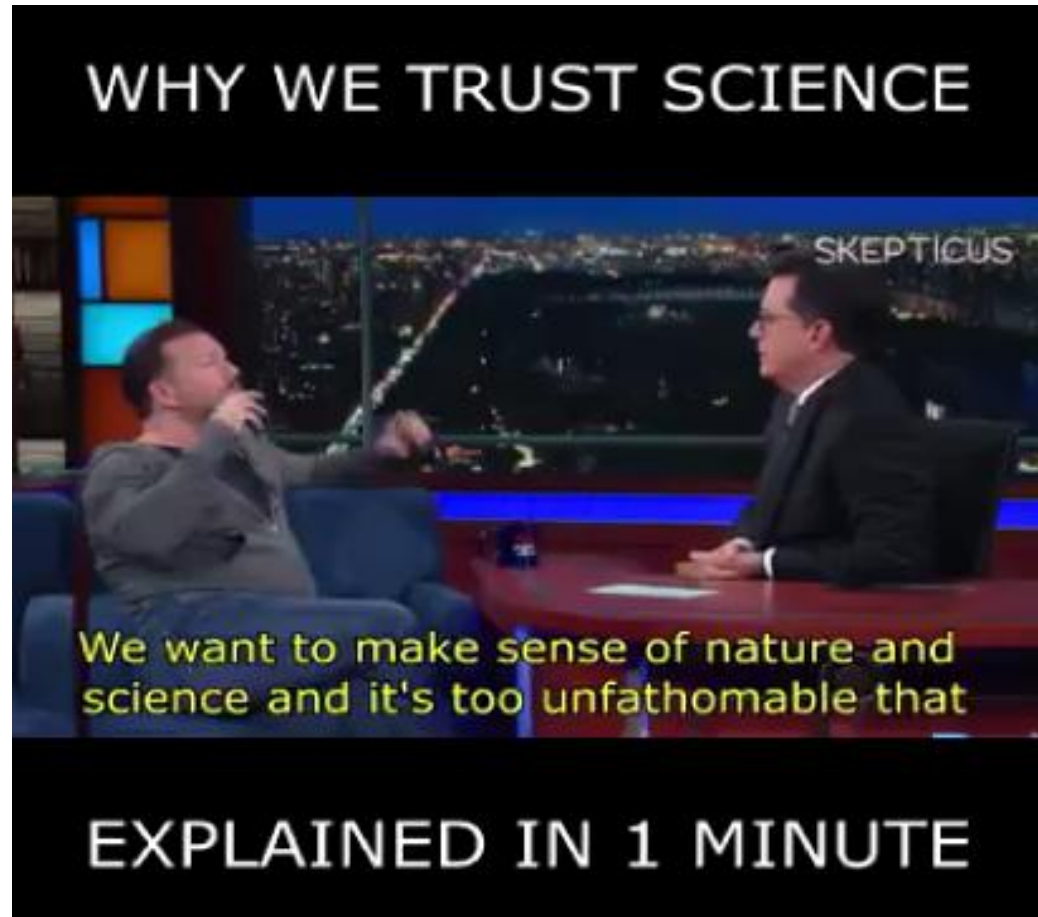
Base: 1,079 Online British adults 16-75 in Sample A, and 1,079 in Sample B. 2-4 February 2024
[Get the data](#) • [Embed](#) • [Download image](#) • Created with [Datawrapper](#)

<https://x.com/keiranpedley/status/1759970566776197307?s=20>

GUBA 1990: 3 DIFFERENT APPROACHES TO THE ONTOLOGY / EPISTEMOLOGY/METHODOLOGY CHALLENGES PRESENTED

	Positivism	Postpositivism (≈ Popper)	Constructivism
Ontology	An objective reality exists, independent of human perception. Can be uncovered (≈ an atom)	An objective reality exists, but it can never be fully known.	Relativist. Multiple (local, context-specific) realities exist as mental constructions, dependent on the persons who hold them
Epistemology	Objective knowledge, that reflects truth. Possible to remove bias and see the world as it is. Aim: Prediction and control	Modified objectivity, because always interaction between inquiry / inquired into. Bias cannot be fully removed. Aim: Prediction and control but fallible.	Subjectivistic, theories emerge from inquiry, not discovered. Inquirer and inquired-into are “fused”, not just interacting. No foundational way of choosing between theories. Aim: Reconstruct meaning (not prediction and control).
Methodology (or investigator)	Methods that can measure the world objectively. Experimental methods preferred. ‘Looking through a microscope’.	Pluralism. Multiple methods and data, both qualitative and quantitative, to reduce biases. Inquiry acknowledges human limitations. ‘See with microscope’.	Hermeneutic and dialectic approach – back and forth. Understand variety of constructions and compare/contrast. Inquiry (re)constructs rather than discovers reality.

RICKY GERVAIS ON (POSITIVISM) SCIENCE



<https://www.facebook.com/scienceteeshirts/videos/1915067131841719/>

GUBA ON POSITIVISM, SOME RELEVANT QUOTES

P. 19: "The basic belief system of positivism is rooted in a *realist* ontology, that is, the belief that there exists a reality *out there*, driven by immutable natural laws. The business of science is to discover the "true" nature" - **ontology**

P. 19: "If there is a real world operating according to natural laws, then the inquirer must behave in ways that put questions directly to nature and allow nature to answer back directly" - **epistemology**

p. 19: "But how can that be done, given the possibility of inquirer bias, on the one hand, and nature's propensity to confound, on the other? The positivist's answer: by the use of manipulative methodology that controls for both... The most appropriate methodology is thus *empirical experimentalism*, or as close an approximation there to as can be managed." - **methodology**

	Positivism
Ontology	An objective reality exists, independent of human perception. Can be uncovered ($\approx$ an atom)
Epistemology	Objective knowledge, that reflects truth. Possible to remove bias and see the world as it is. Aim: Prediction and control
Methodology (or investigator)	Methods that can measure the world objectively. Experimental methods preferred. 'Looking through a microscope'.



Same answer

GUBA ON POST-POSITIVISM, SOME RELEVANT QUOTES AND INSIGHTS

P. 20: "...although a real world driven by real natural causes exist, it is impossible for humans truly to perceive it with their imperfect sensory and intellectual mechanisms" - **ontology**

P. 26: "Even post-positivists have conceded that objectivity is not possible, the results of an inquiry are always shaped by the interaction of inquirer and inquired into". - **epistemology**

P. 21: "Further, if objectivity can never be entirely attained, relying on many different sources makes it less likely that distorted interpretations will be made." - **methodology**

'Modified objectivity' / bias can't be eliminated / approximate truths (≈ Popper's position)

We can only compare validity of a study with another study...never direct access to the world itself

Postpositivism (≈ Popper)

An objective reality exists, but it can never be fully known.

Modified objectivity, because always interaction between inquiry / inquired into.

Bias cannot be fully removed.
Aim: Prediction and control but fallible.

Pluralism. Uses multiple methods and data, both qualitative and quantitative, to reduce biases. Inquiry acknowledges human limitations. 'See with microscope'.



Not necessarily exact same answer

SEE 'THROUGH' (POSITIVISM) OR 'WITH' MICROSCOPE (POST-POSITIVISM) METAPHOR

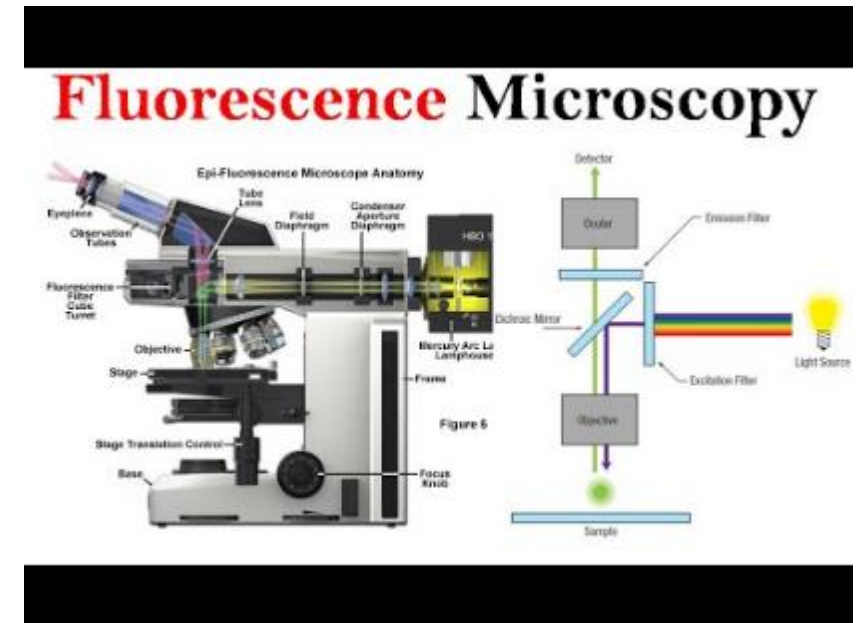


- Positivism: Seeing through the microscope instrument (reality magnified)

- Post-positivism: Modern microscope instrument that chemically alters what is looked at (reality filtered, enhanced & interpreted)

We use survey-instruments

I see myself as someone who...	
1. ...Is talkative	Strongly Disagree 1 2 3 4 5 Strongly Agree
2. ...Tends to find fault with others	Strongly Disagree 1 2 3 4 5 Strongly Agree
3. ...Does a thorough job	Strongly Disagree 1 2 3 4 5 Strongly Agree



POPPER ON TRUTH (EPISTEMOLOGY)



<https://youtu.be/li0ciaqJ0m0?t=155>

GUBA ON CONSTRUCTIVISM, SOME RELEVANT QUOTES AND INSIGHTS

P. 25: "...facts are only within some theoretical framework...basis for discovering "how things really are"...is lost".

P. 26: "Realities are multiple". You can't step outside your theoretical, constructed framework - **ontology**

P. 26: "...the results of an inquiry are always shaped by the interaction of inquirer and inquired into."

p. 25: "...no unequivocal explanation is ever possible. There can be many constructions, and there is no foundational way to choose among them."

P. 26: "...it makes the findings of an inquiry not a report about what is "out there" but the residue of a process that *literally creates them* [the findings]" - **epistemology**

P. 26: " If realities exist only in respondents' minds, subjective interaction seems to be the only way to access them.." **methodology**

Constructivism

Relativist. Multiple (local, context-specific) realities exist as mental constructions, dependent on the persons who hold them

Subjectivistic, theories emerge from inquiry, not discovered. Inquirer and inquired-into are "fused", not just interacting. No foundational way of choosing between theories.

Aim: Reconstruct meaning (not prediction and control).

Hermeneutic and dialectic approach – back and forth. Understand variety of constructions and compare/contrast. Inquiry (re)constructs rather than discovers reality.



INVESTIGATING REMOTE WORK – POST-POSITIVISM VS CONSTRUCTIVISM I

CONSTRUCTIVISM: Multiple, Intangible Mental Constructions (Guba's Ontology)



POSTPOSITIVISM: One Objective Reality, Imperfectly Apprehendable



INVESTIGATING REMOTE WORK – POST-POSITIVISM VS CONSTRUCTIVISM I

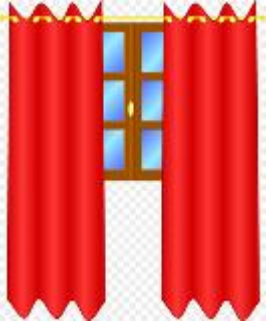
- **Positivist view:** Remote work effect is **objective** and measurable.
 - **Reality:** Real mechanism and effects can be uncovered.
 - **Methods: Quantitative** (experiments, surveys) to find general patterns.
- **Post-positivist view:** Remote work effect is **real** but **bias** and **measurement uncertainty** limits
 - **Reality:** Exists, but can only be **approximated**
 - **Methods: Multiple** / triangulation methods (quant + qual) to reduce **bias** and improve validity.
- **Constructivist view:** Remote work effectiveness **does not exist outside** of the way we enact, talk about and construct it (**differently in different cultures**, industries; tech vs. military e.g.)
 - **Realities:** Multiple realities, depending on perspective (manager, employee, family situation etc.)
 - **Methods: Qualitative** (interviews, observation, in-depth) to study how people make sense of the new management enacted remote work policy – see the world through their eyes

REFLECT: DO THESE 'ISMS' MAKE SENSE TO YOU? (2 MIN)

QUESTIONS?

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DIFFERENT APPROACHES TO INTERVIEWS I (METHODOLOGY)



What kind of approach is this?

digital audio recordings during breaks. To minimize every hour, with occasional ten-minute bathroom breaks in the interim. To maintain the freshness and purity of observations, as well as to make the most of the time provided by short breaks, the embeds recorded their observations verbally, and I then

In a one-month preliminary visit to PrecisionMobile China, I adopted a special methodology to avoid contaminating the environment and the behaviors I was attempting to observe. My research team included three undergraduate students (two females and one male), all three of whom had been born and raised in China until at least the age of ten. The three students were inconspicuously placed on the factory lines as ordinary employees—only the GM, head of HR,

DIFFERENT APPROACHES TO INTERVIEWS II (METHODOLOGY)



What kind of approach is this?

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Shift of RQ and aim: From “does privacy raise performance?” to “how do actors construct and negotiate transparency/privacy?”

Interview approach

- probably not hide identity – use (Harvard, US) identity
- not focus on ‘contamination’ and ‘purity’
- multiple local constructions of transparency/privacy
- Interview, interviews, interpret, and return/contrast interpretations among participants (dialectic analysis, Guba p. 26-27)
- important to ‘see the world through respondents’ eyes’?
- understanding, not causal logic

CHATBOT ANSWER: “CAN YOU GIVE EXAMPLES OF HOW INTERVIEW QUESTIONS WOULD DIFFER IN A POST-POSITIVISTIC VS CONSTRUCTIVIST APPROACH?”

Certainly! Here are examples of interview questions that might differ between a post-positivist and constructivist approach:

Post-positivist approach: In a post-positivist approach, researchers attempt to identify patterns and relationships while acknowledging the limitations of human knowledge. They strive for objectivity and often try to minimize their own biases. Interview questions in a post-positivist approach might include:

Can you describe the steps involved in your decision-making process?

What factors do you consider when evaluating a new opportunity or project?

How do you measure the success of a particular initiative or project?

Constructivist approach: In a constructivist approach, researchers focus on understanding the subjective experiences and meaning-making processes of the participants. They embrace the co-construction of knowledge between the researcher and the participant and acknowledge their own role in shaping the research findings. Interview questions in a constructivist approach might include:

Can you tell me about a time when you faced a difficult decision, and how you made sense of the situation?

How do your personal values and beliefs influence your evaluation of new opportunities or projects?

In what ways has your understanding of success evolved throughout your experiences with various initiatives or projects?

The main difference between the questions lies in the emphasis on the participant's subjective experiences and meaning-making processes in the constructivist approach. While both approaches use interviews to explore participants' perspectives, the constructivist approach places more emphasis on the individual's interpretation of events and personal context. In contrast, the post-positivist approach focuses more on identifying patterns and relationships across participants' experiences, while still acknowledging the limitations of human knowledge and the potential influence of context.”

POST-POSITIVISM VS CONSTRUCTIVISM: DEATH – BIOLOGICAL REALITY OR CONSTRUCTION?

The Post-Positivist View: Death is treated as an objective, physical condition - measurable

The Constructivist Shift:

- Criteria for death involve multiple realities (doctors, families, legal system)
- You can't just declare someone "not-dead" – if realities (societal norms and practices) declare death

Example: When is someone braindead?

Multiple realities – not just one fact



Someone Dead
(Constructed Criteria)



Someone Revived
(Boundary Work)



Someone Maybe Braindead
(Multiple Realities)

POST-POSITIVISM VS CONSTRUCTIVISM: HOW TO STUDY A FLAT EARTH CULT



Post-positivism

- **Ontology:** Earth is (independently) not flat; the belief is false about the physical world.
- **Epistemology:** Researcher aims to stay “outside” and measure belief as a variable.
- **Method:** Explain/predict causes and effects of the belief (e.g., what information exposure, trust, identity factors lead to it; what interventions reduce it)

Post-positivist: False belief to be explained.

Constructivism

- **Ontology:** Multiple local “realities” of meaning—flat-earth is a lived, socially maintained reality for members.
- **Epistemology:** Findings are co-created in interaction
- **Method:** Compare constructions across stakeholders (believers, skeptics, media, schools, platforms); examine who defines, performs and contest “evidence,” “science,” and “authority” etc.

Constructivist: Real social world to be understood.

POST-POSITIVISM VS CONSTRUCTIVISM: A FEW NOTES

Typical (PP) aim: Find out what is really the case

Most (?) bachelor projects: Some variant of post-positivism (PP)

- What is aim: To find information (like a treasure-hunt/archeologist) – or understand how realities are (re)constructed?
- Interfering: Aiming to be as invisible as possible – or can interviewer react, be passionate (e.g. in interview situation)?
- Interviewguide: Standardized – or evolving?
- Follow-up logic: Qual → hypothesis testing? – or a study in its own right
- How do you assess objectivity? 'Truth' – or trustworthiness/authenticity
-



Not always easy to differentiate PP and C!

WHAT POSITION DO (PSYCHOLOGY) RESEARCHERS HAVE?



Tim van der Zee

@Research_Tim

Follow



How do you see latent variables such as "motivation" or "extraversion"; do they really exist? Are they 'unreal' but useful descriptions? Does it even matter?

31% Real

17% Not real

20% Cant know/doesnt matter

32% See results

445 votes • Final results



“The dialog is not to determine which paradigm is, finally, to win out” Guba 1990, p. 27

https://twitter.com/research_tim/status/993756255423430656?s=11

WHERE DOES KUHN FIT INTO GUBA'S FRAMEWORK?

VEVOX.APP 143-136-950 (CLICK ON IMAGE)

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KUHN AND GUBA

“However, the classification of Kuhn into the paradigm of constructivism is not perfect. A central claim of constructivism is that due to the undetermination of theory, i.e. that *“no theory can ever be fully tested”, “no unequivocal explanation is ever possible. There can be many constructions, and there is no foundational way to choose among them.”* (Guba 1990, pp. 25) This stands somewhat in contrast to Kuhn’s perspectives on scientific development. Indeed, while he argues that the incommensurability of paradigms implies that both may be right because they cannot be compared; *“The proponents of different theories are like the members of different language-culture communities. Recognizing the parallelism suggests that in some sense both groups may be right.”* (Kuhn 1962, pp. 205) However, Kuhn does then go on to argue – somewhat – that scientific progress means later theories are better for solving scientific “puzzles” than earlier theories; *“...scientific development is, like biological, a unidirectional and irreversible process. Later scientific theories are better than earlier ones for solving puzzles in the often quite different environments to which they are applied.”* (Kuhn 1962, pp. 206) This implies that Kuhn is not entirely aligned with the subjectivist epistemology of constructivism despite seemingly being aligned with the relativist ontology of the paradigm. Thus, it seems that Kuhn does not adhere perfectly to any of the three paradigms defined by Guba (1990), but that his perspectives on ontology places him closest to the constructivist view on the social science world.”

(from a student exam answer)

NELSON 2016 AND PARADIGMS

(REFLECT 2 MIN)

Question 5 (21%) – 2020 Spring exam

How would you categorise Nelson's (2016) take on business economics as a science: Is it a positivist, post-positivist or constructivist position, as per Guba's (1990) categorisation of these positions? Please be specific in your answer and offer at least two quotes from Nelson (2016) that support your argument.

NELSON 2016, SOME QUOTES

Clearly not a positivist. Post-positivist or constructivist, according to Guba?

“That is, the number, and the assumptions behind its construction, comes to be treated as what the phenomenon is, rather than as a useful if limited measure of that phenomenon.” (Nelson 2016, p. 1695).

“The numbers used in the social and behavioral sciences almost always are, by themselves, somewhat limited and imprecise characterizers of the phenomena they are designed to measure, and need to be understood as a part of a broader and more detailed if qualitative characterization (Nelson 2016, p. 1695).

“And in these fields the kind of precise law-like relationships that physics has identified simply may not exist, even in the sense of tightly stable stochastic relationships.” (Nelson 2016, p. 1693)

“If these phenomena are to be associated with numbers, some proxies need to be invented or some quantitative indicators constructed that relate to phenomena of interest, but are not natural measures of that.” (Nelson 2016, p. 1696)

Guba 1990, p.26: “First, it renders the distinction between ontology and epistemology obsolete; what can be known and the individual who comes to know it are fused into a coherent whole””

A GOOD ANSWER TO 2020 Q5, PART 1

To categorise Nelson's take on business economics as a science as per Guba (1990), I will need to first establish what Guba's positions are, as to be able to place Nelson in one (or several) of them. In his 1990 paper titled "The Alternative Paradigm Dialog", Guba discusses the three paradigms of positivism, post-positivism and constructionism. He discusses how the different paradigms will have different answers to the ontological, epistemological and methodological questions (Guba 1990, page 18).

Positivism has the ontology that concepts and theories are real, i.e. there is one real "truth". The epistemology is that the theories used then represents the world as it is, and is therefore objective knowledge.

Post-positivism has almost the same ontology as positivism, in that there is one real "truth". However, they are more critical about one's access to it, as they argue it is not reachable. Their epistemology is that objectivity then is the ideal, but we will never be able to see the world as it is, and therefore our best bet is theories that approximate the real "truth" as closely as possible.

Constructivism has a completely different ontology, as this paradigm argues that there is several "truths". As Guba puts it: " "Reality" exists only in the context of a mental framework (construct) for thinking about it" (Guba 1990, page 25). This also means, in relation to constructivism's epistemology, that theories are constructed, and that the constructed reality only then can be "seen" through this window of constructed theory (Guba 1990, page 25).

At first, I establish why I do NOT consider Nelson to be a positivist. As described above, a positivist will argue that the theories they use are a window straight into how the world really is, and is therefore objective knowledge. However, in his paper, Nelson several times mentions that many variables in social science does not make sense to evaluate at face value: "A good part of the reason why one or a few numbers alone generally do not cover adequately the subject matter being studied, and the numbers themselves often are somewhat fuzzy, is that the phenomena studied have several aspects and each of these has blurry edges." (Nelson 2016, page 1696). Nelson argues that we cannot just use one number to define a variable such as unemployment, as unemployment as a subject is inherently fuzzy. You can get different arguably correct answers to what one considers unemployment. Whether it be people out of work, people out of work but who are willing to work etc. But you cannot get different answers to what the weight of a specific atom is (obviously given the respondent knows the answer). And therefore, it does not make sense to view numbers or theories in the social science field as a window into reality, as there needs to be accounted for their inherent fuzziness.

A GOOD ANSWER TO 2020 Q5, PART II

A quote that could place Nelson on the more constructivist side is the following: “Many of the particular subject matters treated by the social and behavioural sciences not only are quite heterogeneous, but the general conception of the nature of the phenomenon often has uncertain boundaries.” (Nelson, page 1695). In the quote, Nelson argues that the nature of phenomena has uncertain boundaries, which I think hints at him meaning the phenomena are a construct of the created data and the given researcher, and should thus be treated as such. However, one could also argue that he says the uncertain boundaries means that the data are only an approximation to reality, which would then seem like him being a post-positivist

In general, I would argue the way he addresses quantifiable measures makes him seem like a post-positivist, as it seems as though some data/numbers do have a face value on their own, such as in the following quote: “The numbers used in the social and behavioural sciences almost always are, by themselves, somewhat limited and imprecise characterizers of the phenomena they are designed to measure, and need to be understood as part of a broader and more detailed if qualitative characterization.” (Nelson 2016, page 1695). However, the way he addresses that data needs to be understood as part of a broader qualitative characterization makes him seem more constructivist, as it seems he relies more on the creation between the data and the researcher to create meaning.

To conclude, I would argue that Nelson’s 2016 paper is definitely not positivistic, as he does not believe theories are a window of the one “true” reality. However, when evaluating whether he is post-positivistic or constructivist, I would argue that the discussion gets blurrier, and that he is somewhere in between.

ANOTHER EXAMPLE: STUDYING THE VARYING KINDS OF IMPACT OF BREASTFEEDING I

Example presented in this video from 2020

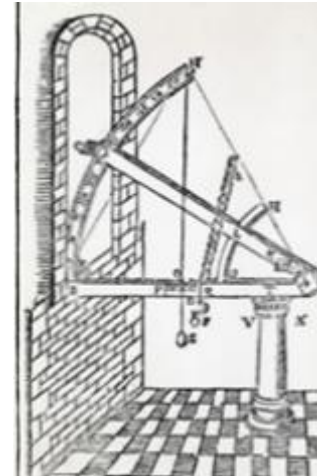
<https://youtu.be/4ZOeQomRHpc>



STUDYING THE VARYING KINDS OF IMPACT OF BREASTFEEDING II (BIOLOGICAL EFFECTS)



<https://w0.pngwave.com/png/791/656/medical-laboratory-hospital-physician-science-science-png-clip-art.png>



BREASTMILK IS GOOD FOR THE BABY! (BIOLOGICAL EFFECTS)



Breast milk ----- >

Good for baby



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BUT...IT IS NOT SO SIMPLE

WHAT IS THE RESEARCH QUESTION?

Breastfeeding can be very painful, and basically impossible → potentially loss of sleep & **self-esteem**, + **feeling of shame** by mother

How do we (people and researchers) construct the meaning of these concepts?

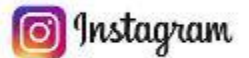


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- Norms of self-esteem and when to feel shame vary across cultures ≠ physiological impact
- Maybe 'level of esteem/shame' aren't independent reality?
- Mothers' emotions (shame, self-esteem) can also → wellbeing of baby



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BREASTMILK: POST-POSITIVISM VS CONSTRUCTIVISM

Post-positivist Approach:

- Biological effects of breastmilk **independent** of researchers / culture
- Maternal **pain levels** and **self-esteem & baby well being** can be measured **quantitatively**



Constructivist Approach:

- **No independent** effects of breastmilk
- How do **mothers, doctors, and policymakers construct** the meaning of "good motherhood" and **pain, self-esteem and shame** – in different contexts?
- What **social and cultural expectations** shape the experience of breastfeeding?

WHAT IS THE RESEARCH QUESTIONS (RQ) ↔ PHILOSOPHY OF SCIENCE ASSUMPTIONS

When RQ shifts from physiological components to psychological considerations

- Shift in how distinct, pre-existing the reality is?
- Shift in how certain we can be about actually measuring what is out there?
- Might the shift require moving from measurement to engaging in more reflexive conversation with anxious mothers?

What paradigm applies – thus also depends on the RQ and the phenomena the RQ seeks to investigate

Also: If one has certain (paradigm) assumptions, one addresses certain RQs



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GUBA'S PARADIGMS AND METAPHORS OF THEORY

(CF. LESSON 6)

Philosophical assumptions: The causality approach is often associated with positivism or post-positivism—focusing on objective data and measurable outcomes. In contrast, the understanding approach is more closely aligned with constructivism, which values the subjective constructions of reality as experienced by individuals, therefore emphasizing context and meaning over general explanations (cf. Guba 1990, curriculum in a later lesson). These differing philosophical assumptions shape how researchers frame their questions and interpret their finding (cf. Guba 1990, curriculum in a later lesson).

Bergenholtz 2026f: p. 8-9, section 2.3

Causal Logic	Understanding
Incentives → Motivation → Effort → Performance (counterfactual reasoning)	Motivation is enacted differently depending on context and prior experiences
Direct measurement (surveys, experiments)	Meaning is constructed through interaction, dialogue, and shared reflection
Assumes stable cause-effect relationships	Understanding develops through interpretation and discussion

HOW DO WE ASSESS CAUSALITY / TRUTH / QUALITY OF WORK, IN DIFFERENT ISMS? I

Causal logic (associated with)

- Positivism: Truth (validity, hitting the bullseye exactly)
- Post-positivism: Approximating truth (validity, closer to bullseye)



Understanding (associated with)

- Constructivism: “Constructions are not more or less "true," in any absolute sense, but simply more or less informed and / or sophisticated.” (Lincoln & Guba 1994, p. 110)

→ quality criteria: credibility, authenticity, transparency

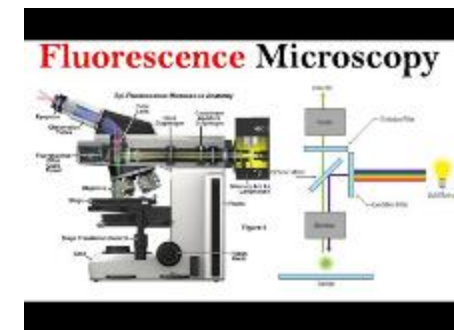


SUMMING UP LESSON 9

- (1) *Ontological*: What is the nature of the “knowable”? Or, what is the nature of “reality”?
- (2) *Epistemological*: What is the nature of the relationship between the knower (the inquirer) and the known (or knowable)?
- (3) *Methodological*: How should the inquirer go about finding out knowledge?



	Positivism	Postpositivism (≈ Popper)	Constructivism
Ontology	An objective reality exists, independent of human perception. Can be uncovered (≈ an atom)	An objective reality exists, but it can never be fully known.	Relativist. Multiple (local, context-specific) realities exist as mental constructions, dependent on the persons who hold them
Epistemology	Objective knowledge, that reflects truth. Possible to remove bias and see the world as it is. Aim: Prediction and control	Modified objectivity, because always interaction between inquiry / inquired into. Bias cannot be fully removed. Aim: Prediction and control but fallible.	Subjectivistic, theories emerge from inquiry, not discovered. Inquirer and inquired-into are “fused”, not just interacting. No foundational way of choosing between theories. Aim: Reconstruct meaning (not prediction and control).
Methodology (or investigator)	Methods that can measure the world objectively. Experimental methods preferred. ‘Looking through a microscope’.	Pluralism. Uses multiple methods and data, both qualitative and quantitative, to reduce biases. Inquiry acknowledges human limitations. ‘See with microscope’.	Hermeneutic and dialectic approach – back and forth. Understand variety of constructions and compare/contrast. Inquiry (re)constructs rather than discovers reality.



WHAT POSITION DO (PSYCHOLOGY) RESEARCHERS HAVE?

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How do you see latent variables such as “motivation” or “extraversion”; do they really exist? Are they ‘unreal’ but useful descriptions? Does it even matter?

31% Real
17% Motivated
39% Can't know/doesn't matter
15% Not sure

443 votes • Final results

<https://twitter.com/timvanderzee/status/1518555535456789123>

“The dialog is not to determine which paradigm is, finally, to win out” Guba 1990, p. 27